

OpenTrace Africa — mart_dev complete entity dictionary, ERD, and indicator-class analytical guide

Dataset: mart_dev

Version: warehouse Aug 2026 (post production–yield split; 63 live models)

Rule: every published number states **grain + source_key + as_of_date + unit**.

This document describes **physical mart tables**, how they join, and how to write insight for each **indicator class** (EL, GYI, FVC, FS, PROD, PRC, CLIM, SOIL, AH, VEG, ENV, INP, HDI, BIO, RES).

Related docs

Doc	Role
OpenTrace Mart Complete Guide.pdf	Printable PDF of this guide (ERDs as images)
OpenTrace Mart Entity Dictionary.xlsx	Branded catalogue + indicator-class map + full Columns (with descriptions)
mart_dev_entity_dictionary.xlsx	Engineer dump (Entities / Columns / Relationships / Recipes / ACF)
MART DEV OTA ANALYST GUIDE.docx	OTA report-writing playbook
CATALOG TO MART MAP.md	Catalog ↔ mart glossary + ACF contract
MART QA NOTES.md	QA inventory and known caveats

1. How to use the relational database

```
raw_dev → staging_dev → intermediate_dev → mart_dev
                                     |— dim_*   (22)
                                     |— fct_*   (27)
                                     |— agg_*   (11)
                                     |— bridge_* (3)
```

```
SELECT
  g.country_name,
  f.as_of_date,
  -- measures
  s.organisation_name,
  s.tier,
  f.source_natural_key
FROM `project.mart_dev.fct_...` f
LEFT JOIN `project.mart_dev.dim_geography` g ON f.geography_key = g.geography_key
LEFT JOIN `project.mart_dev.dim_source`    s ON f.source_key     = s.source_key
WHERE f.as_of_date BETWEEN DATE '2020-01-01' AND DATE '2024-12-31'
      AND g.country_iso3 = 'ETH'
```

Filter `as_of_date` with a **range** (partition prune). Do not wrap it in `EXTRACT(YEAR ...)`.

Hard rules

- 1. Do **not** mix FAOSTAT country–year (fct_production) with FNID–season (fct_yield). They are **separate facts**.
- 2. On fct_production , filter production_grain (physical | index | gross_value) before area/qty charts; use physical for agg_production_country_year .
- 3. Do not mix IPC measure_type population with classification.
- 4. Do not mix economics measurement_form (share / current / constant / growth).
- 5. Do not average iSDA and ISRIC.
- 6. Do not present ILRI microdata as national official statistics.
- 7. Do not blend FEWS / WFP / FAOSTAT prices without keeping source_key (and price type/source attributes).
- 8. On fct_trade , filter trade_grain (faostat_country_year | fews_border_month) before blending.
- 9. loaded_at is not ACF freshness unless as_of_date_basis = 'loaded_at' .
- 10. No published fact row without source_key .
- 11. Off-season / décrue / dam ≠ rainfed main season.
- 12. Child stunting/wasting and labelled regenerative adoption remain **out of scope**.
- 13. ILRI dairy cow-day milk/calving stays **out** of production/yield facts by design.

ACF / RAG

Cite **rows used in the answer**. Map:

ACF	Warehouse
tier	dim_source.tier (or fact tier when present) — 1 global, 2 national, 3 community
data_level	fact data_level (warehouse row resolution)
as_of_date	fact as_of_date (+ note as_of_date_basis)
geo_scope	fact geo_scope / place_scope (names/labels) — not a dim_geography column
metric	fact metric or product / indicator / element identity
magnitude, unit	value + unit
source_id	source_key / source_id

2. Entity catalogue — dimensions

Table	Holds	Grain	PK
dim_geography	Country, admin1/2, city, FNID, lat/lon, population; data_level1	One place	geography_key
dim_ref_country	ISO3 reference + Africa scope policy	One ISO3	country_iso3
dim_faostat_area	FAOSTAT area_code → ISO/M49 attrs	One area_code	faostat_area_key
dim_date	Calendar day → year/month/quarter	One day	date_key
dim_season	Country-specific season, plant/harvest months, off-season	Country + season	season_key
dim_source	Producer, ACF tier, default data_level, lineage	One dataset	source_key
dim_product	Crops/commodities + CPC	Product	product_key
dim_item	FAOSTAT/macro item	Item	item_key
dim_element	FAOSTAT element	Element	element_key
dim_indicator	Named series (employment, UNCCD, ASTI, ClimateWatch)	Indicator	indicator_key
dim_unit	Units and currencies	Code	unit_key
dim_classification	IPC phase + scale	Phase	classification_key
dim_market	Marketplaces	Country + market	market_key
dim_household	Survey HH attributes (not measures)	Source + hh id	household_key
dim_soil_property	Property + depth	Property + depth	soil_property_key
dim_livestock	Species	Species	livestock_key
dim_disease	Foodborne / hazard names	Name	disease_key
dim_aez	AEZ code/name/version	Code + version	aez_key
dim_sex	Male / female / total / unknown	Code	sex_key
dim_organisation	OpenAIRE + ASTI institutions	Org	organisation_key
dim_person	OpenAIRE persons	Person	person_key
dim_research_project	OpenAIRE projects	Project	research_project_key

dim_geography columns

geography_key, geo_level1 (country|admin1|admin2|city|fnid|...), data_level1 (national|sub_national|community|point), country_name, country_iso2, country_iso3, admin_1_name, admin_2_name, city_name, fnid, latitude, longitude, population, capital_status, parent_geography_key / country_key, loaded_at.

ACF **geo_scope** lives on **facts**, not on this dimension.

dim_source columns

source_key, source_natural_key, organisation_name, tier, default_data_level, producer_scale, loaded_at.

Producer scale ≠ row geo_scope ≠ granularity.

dim_season columns

season_key, country, season_name, season_name_norm, start_month, end_month, crosses_year, is_off_season, ...

Join **yield** on season_key (or country + season_name_norm). Do **not** require season on FAOSTAT fct_production.

3. Entity catalogue — facts

Table	Measures	Grain	Discriminator
fct_production	value + element; convenience area_harvested / production_qty / yield_value on physical	FAOSTAT country–year long-form	production_grain (physical index gross_value)
fct_yield	area_harvested, production_qty, yield_value	FNID–season (yield_raw_data)	— (separate SoT from production)
fct_food_security	value, pct_phase3–5	FNID × month × scenario	measure_type
fct_hdi	value (HDI)	country × year	
fct_prices	value, common_unit_price	market × product × type × month × source	source / price_type attrs
fct_trade	value / flow attrs	faostat_country_year or fews_border_month	trade_grain
fct_soil_health	value	point × property × depth × source	source (iSDA ≠ ISRIC)
fct_household	FIES, incomes, food/energy, gender control, diversity	household	project microdata
fct_employment	value	country × indicator × sex × year	unit, sex
fct_economics	value	country × item × element × year	measurement_form
fct_climate	metric + value	point_obs OR country_model	climate_grain
fct_land_inputs	value	use / trade / other	input_grain
fct_emissions	value	country × item × element × year	
fct_food_balance	value + convenience production/imports/exports/food/feed/losses	country × product × element × year	map food/feed/losses by element_code
fct_animal_health	CDS, incidence, mortality, cost	household × species	
fct_food_hazards	samples, prevalence	study × hazard	not farm grain
fct_forestry	value	area × item × year	forestry_grain
fct_humanitarian	value	recipient × item × year	
fct_vegetation	NDVI or carrying_capacity	grid OR site	vegetation_grain
fct_gender_inclusion	value	country × year SDG	keep separate from HH gender
fct_investment	value	country × year + donor/sex	
fct_air_quality	value (e.g. PM)	sensor × timestamp	Nakuru / point
fct_biodiversity	counts / attrs	occurrence	

Table	Measures	Grain	Discriminator
fct_protected_areas	area / presence	feature	
fct_germplasm	accession presence	accession	
fct_machinery	value	country × item × year	
fct_insurance	categorical attrs	household × record_type	

Common fact columns: [fact]_key , geography_key (if spatial), source_key , source_natural_key , as_of_date , as_of_date_basis , ACF fields (tier , data_level , place_scope , metric , source_id , geo_scope), year / month / date_key , measures, unit , loaded_at .

4. Aggregates and bridges

Aggregate	Use
agg_production_country_year	National production scorecard (physical production only)
agg_production_country_season	FNID yield rolled to country × season
agg_food_security_country_month	National IPC snapshot (population measure)
agg_hdi_latest	Latest HDI
agg_prices_country_month	Country-month prices (keep source_key)
agg_soil_admin	Admin mean soil, still split by source
agg_employment_country_year	indicator × sex × unit × source
agg_economics_country_year	item × measurement_form × source
agg_emissions_country_year	item × element × source
agg_food_balance_country_year	product × source
agg_forestry_country_year	forestry_grain × source

Yield in seasonal aggregates = SUM(production_qty) / SUM(area_harvested) from fct_yield .

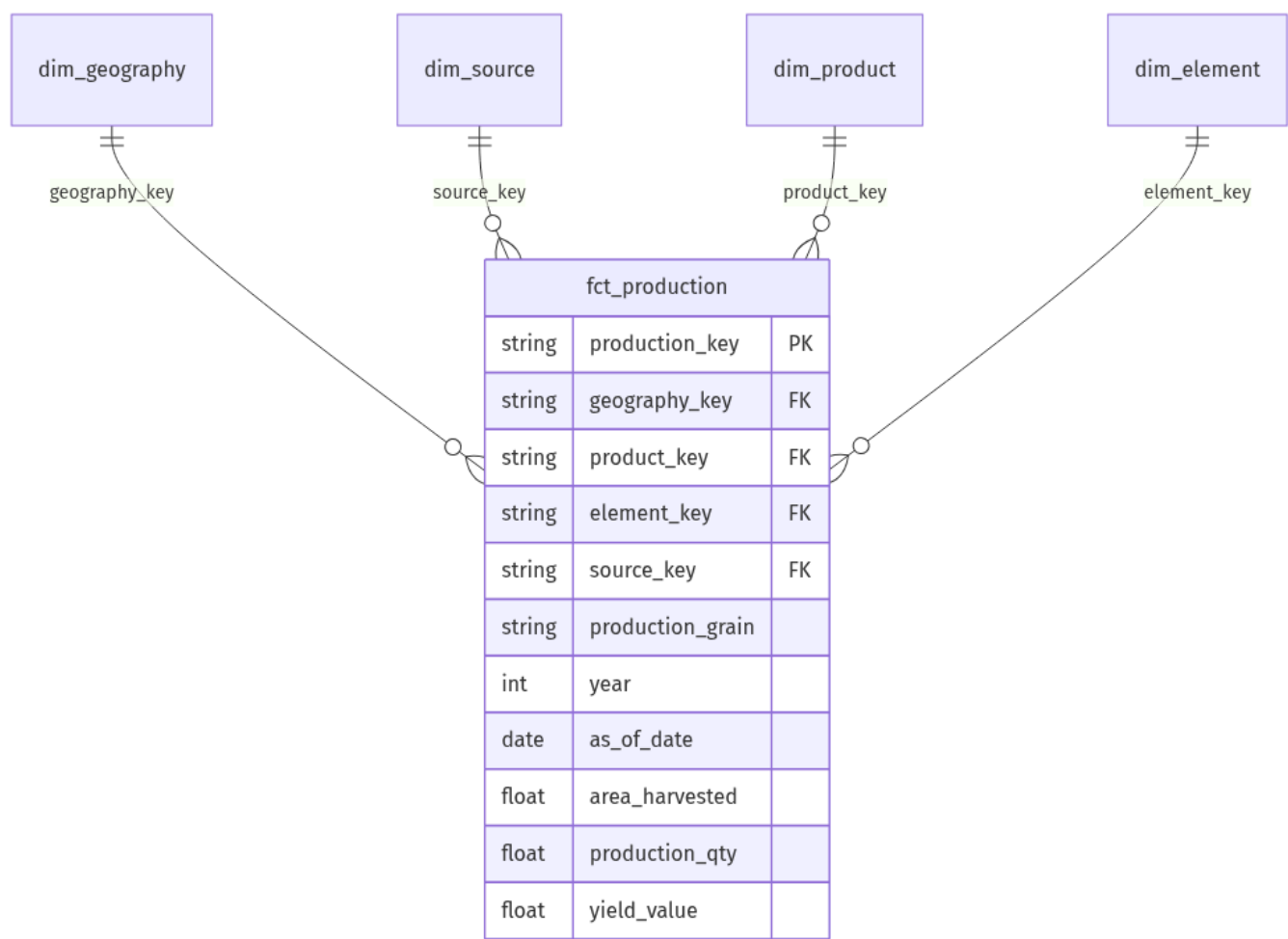
Bridge	Meaning
bridge_geography_aez	place ↔ AEZ
bridge_research_org	OpenAIRE project ↔ organisation
bridge_research_person	OpenAIRE project ↔ person

5. Full schema ERD (holistic)

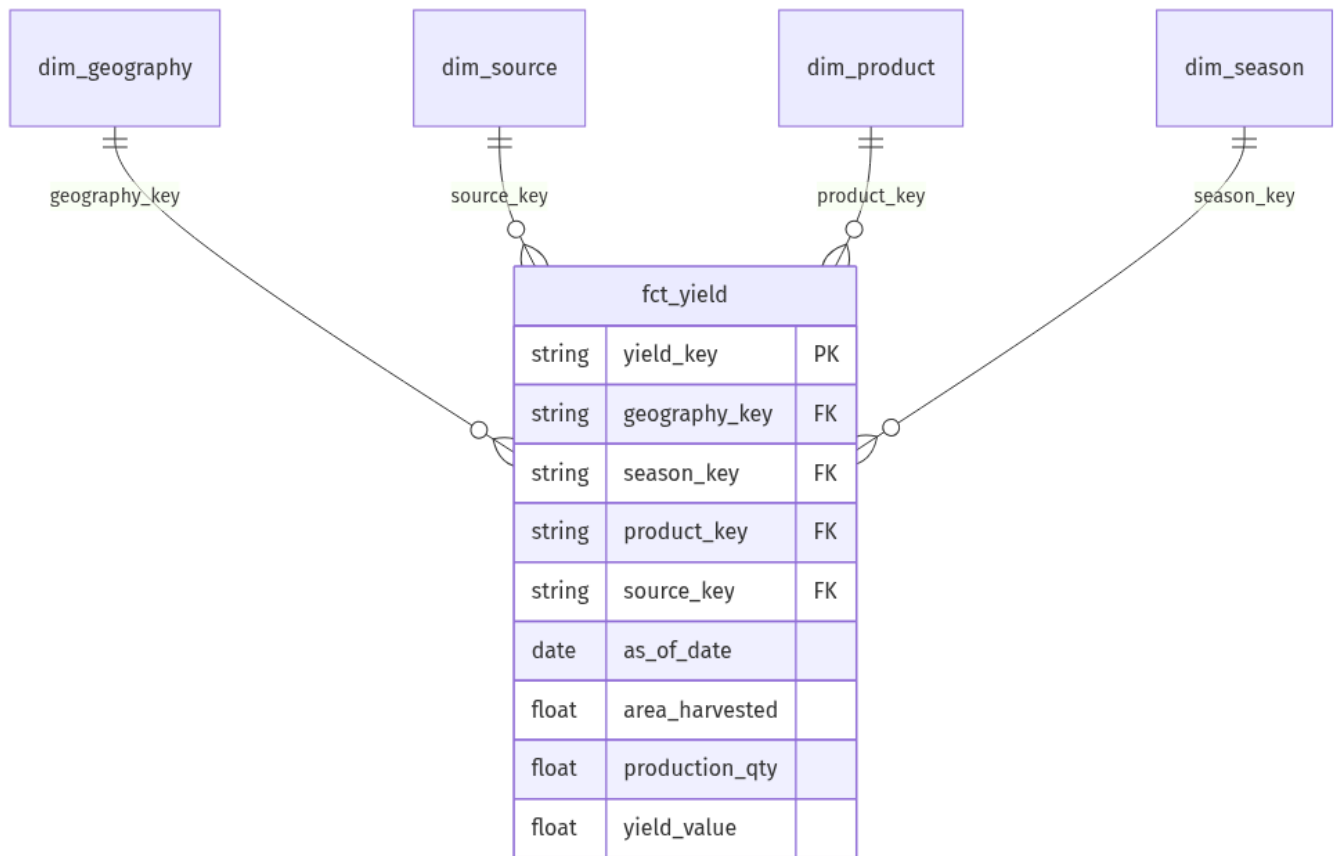


Paste the mermaid into mermaid.live if the renderer caps node count.

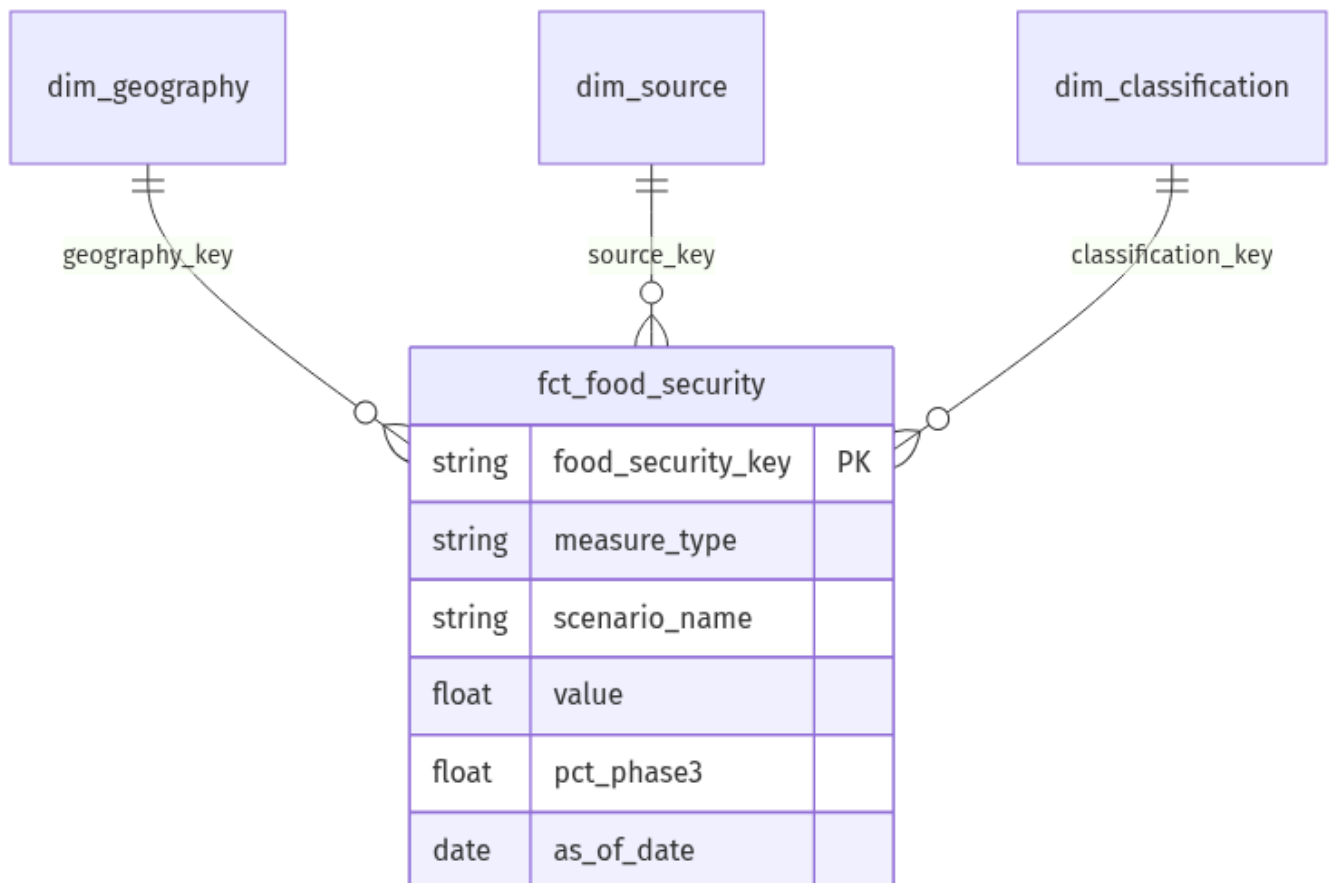
Star: FAOSTAT production (country-year)



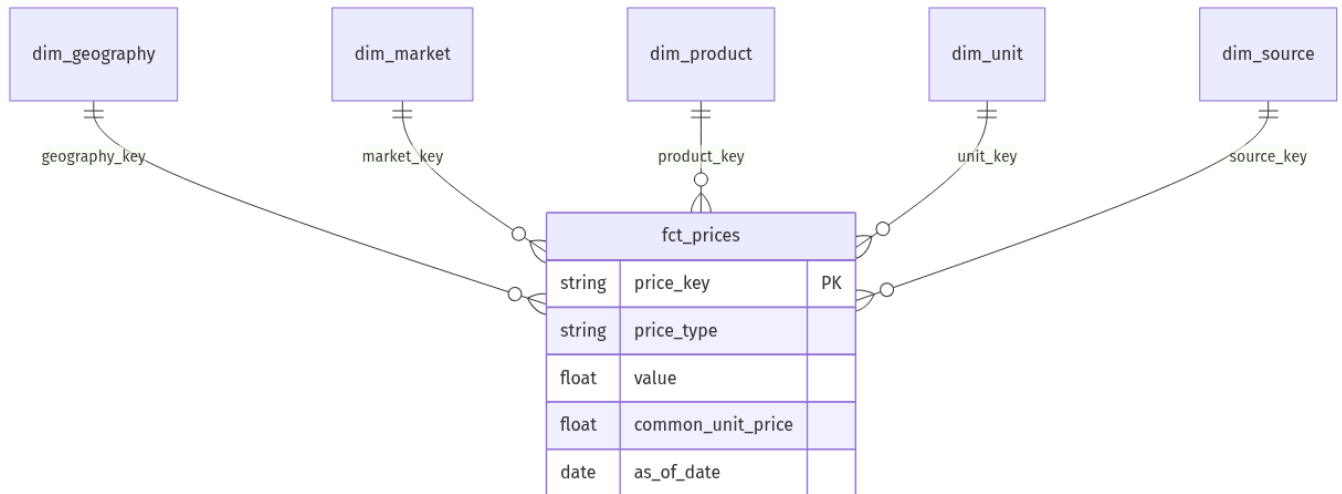
Star: seasonal yield (FNID-season)



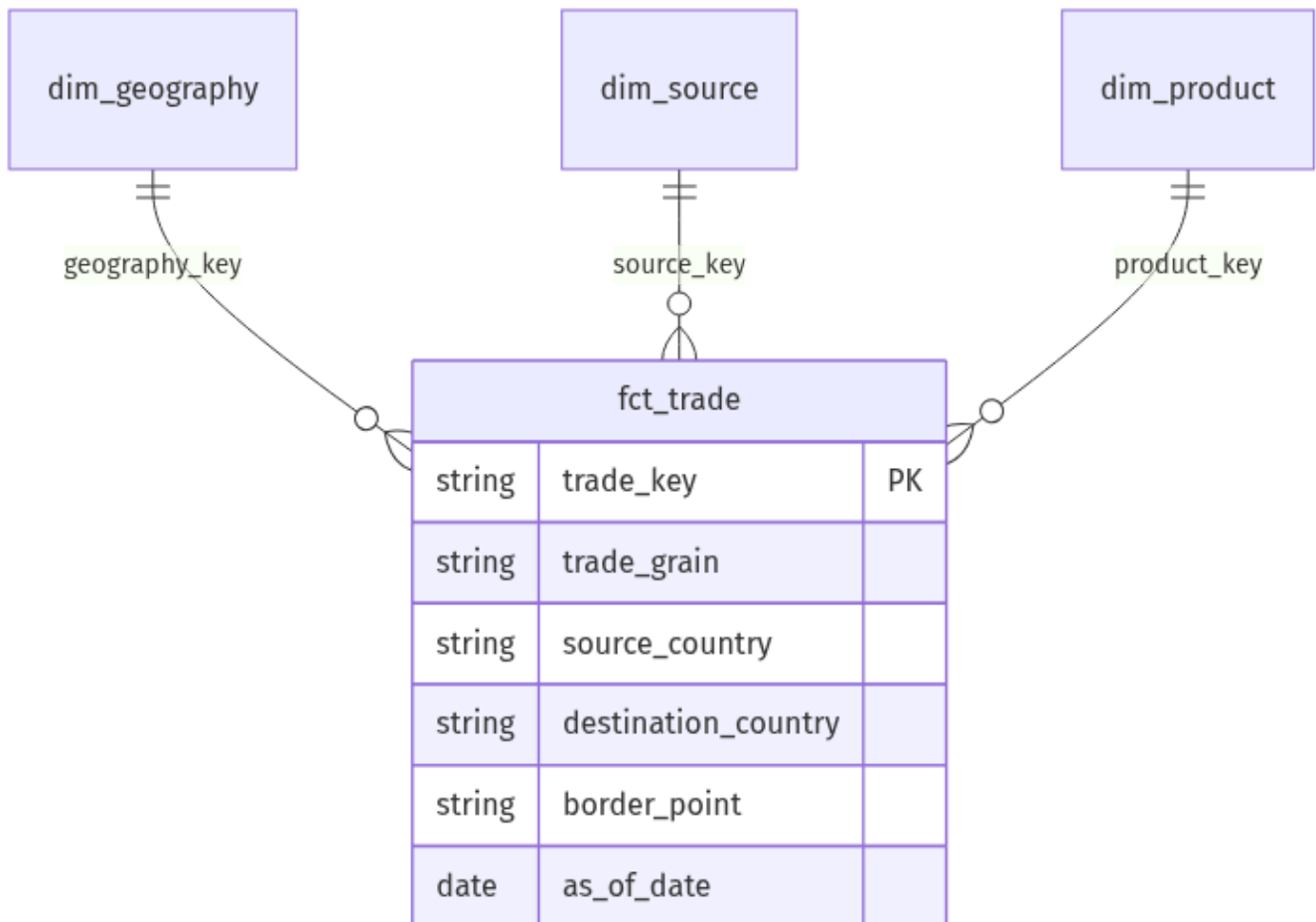
Star: food security



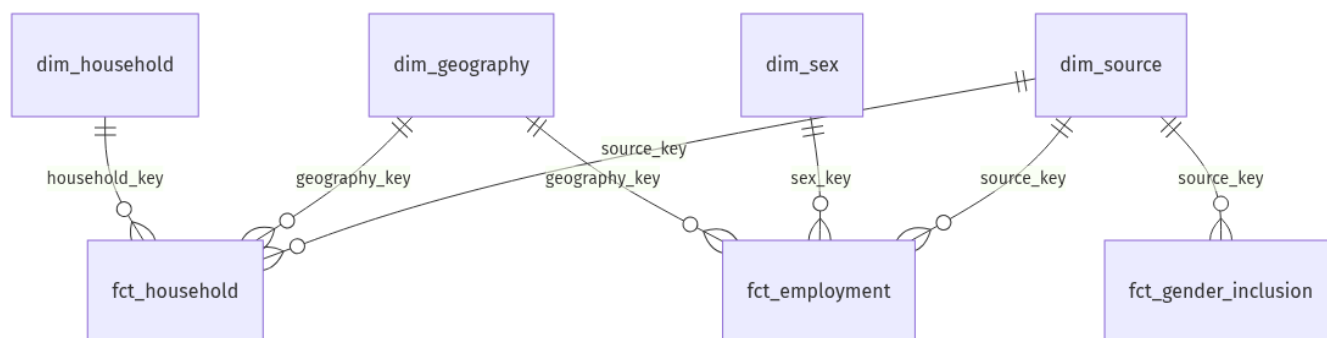
Star: prices



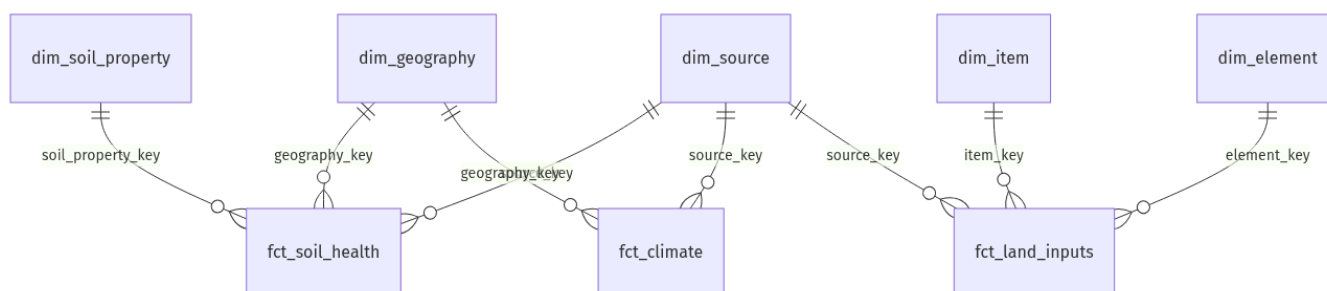
Star: trade



Star: household / gender / employment



Star: soil / climate / land



6. Analytical guide by indicator class

Write insights as: **claim + grain + place + period + source + caveat.**

EL — Economic and livelihood

Facts: `fct_employment`, `fct_economics`, `fct_household`

Families: employment in agriculture; income mix; GDP / value-added context.

Insight type	Query pattern	Do not
Ag employment share	<code>fct_employment</code> + indicator + unit = '%' + sex	Mix % with 1000 persons
Farm vs off-farm income	<code>fct_household</code> farm_income, offfarm_income	Treat sample as census
Aff value added	<code>fct_economics</code> measurement_form = 'share_of_gdp'	Mix with current USD

Example claim: "In Ethiopia in 2020, agricultural value added was X% of GDP (FAOSTAT macro, national year)."

GYI — Gender, youth and inclusion

Facts: `fct_gender_inclusion` (national SDG), `fct_household` (control scores), `fct_employment` by sex.

Never union HH control scores into official SDG land-rights rows. State sample vs official.

FVC — Food system and value chain

Facts: `fct_economics` (value added forms), `fct_food_balance` (losses, food vs feed), `fct_prices`, `fct_trade` (`trade_grain`), `fct_forestry` (`forestry_grain`).

- Filter `fct_trade.trade_grain` before border vs national trade claims.
- Loss insight: `fct_food_balance` losses (via `element_code`) is SUA disappearance, not measured farm-gate waste.

FS — Food security and nutrition

Facts: `fct_food_security`, `fct_household` (FIES, energy, availability), `fct_humanitarian`, `fct_food_balance` food elements.

Claim	Source	Caveat
People in IPC 3+	<code>fct_food_security measure_type=population</code>	Not classification rows
Outlook	<code>scenario_name</code>	Projected ≠ current
Household food insecurity	<code>fct_household fies_score</code>	Project sample
Child stunting	—	Out of scope

PROD — Agricultural production

Facts: `fct_production` (FAOSTAT country–year) **and** `fct_yield` (FNID–season).

Always keep them separate.

Claim type	Table / agg	Filter
National official-style production	<code>fct_production</code> or <code>agg_production_country_year</code>	<code>production_grain = 'physical'</code> for area/qty
Indices / gross value	<code>fct_production</code>	<code>production_grain</code> in (<code>index</code> , <code>gross_value</code>)
Seasonal subnational yield	<code>fct_yield</code> or <code>agg_production_country_season</code>	<code>season_key</code> / harvest year

Yield in seasonal rollups: sum production / sum area from `fct_yield`.

Do **not** write “FAOSTAT Meher FNID yield” from `fct_production`.

PRC — Prices and markets

Fact: `fct_prices`

Keep `source_key` (and price type attributes).

`common_unit_price` is FEWS-oriented; WFP null is expected.

Month must be calendar 1–12 (or null) — not FAOSTAT label codes like 7004.

CLIM — Climate and weather

Fact: `fct_climate`

Filter `climate_grain`: `point_obs` (NASA/ERA5) vs `country_model` (ClimateWatch).

Prefer long `metric` + `value`.

SOIL — Soil health and land

Facts: `fct_soil_health`, `fct_land_inputs` (`input_grain`). Prefer `agg_soil_admin` for admin summaries.

- Compare iSDA vs ISRIC; do not average.
- Land use vs fertilizer trade vs other: filter `input_grain`.
- AEZ via `bridge_geography_aez` when populated.

AH — Animal health

Facts: `fct_animal_health`, `fct_food_hazards`, `fct_insurance`.

Farm CDS ≠ study-level prevalence. Insurance is mostly categorical (herd band, start year).

VEG — Vegetation

Fact: `fct_vegetation`

`vegetation_grain`: `ndvi_grid` vs `ilri_site` (carrying capacity). Do not treat NDVI as yield.

ENV — Environment / emissions

Fact: `fct_emissions`

Filter element (total vs intensity) and keep `source_key`.

INP — Inputs

Same physical table as land: `fct_land_inputs`

Fertilizer / pesticide use vs trade (`input_grain` + item). Also `fct_machinery` for stock/use.

HDI — Human development

Facts: `fct_hdi`, GDP via `fct_economics`.

Country-year only. If `geography_key` / `country_iso3` is null, do not publish a mapped insight until geo resolves (see MART_QA_NOTES). Prefer `agg_hdi_latest` for snapshots.

BIO — Biodiversity and protection

Facts: `fct_biodiversity`, `fct_protected_areas`, `fct_germplasm`.

Occurrence ≠ national biodiversity index. Germplasm `as_of_date` may be `loaded_at` basis.

RES — Research system

Dims/bridges: `dim_research_project`, `dim_organisation`, `dim_person`, `bridge_research_org`, `bridge_research_person`.

ASTI researchers/spend: `fct_investment` ASTI branches (sex, degree, institution).

7. Insight-writing template

Claim: <metric> <direction/magnitude> in <place> during <period>
Grain: <country-year | fnid-season | household | point>
Source: <organisation> (source_natural_key / tier)
Unit: <unit>
Caveats: <grain mix avoided; sample vs official; expected nulls>
ACF inputs present: tier, data_level, as_of_date, geo_scope/place_scope, metric, source_id

Worked examples

PROD (seasonal): "Maize yield in Ethiopia Meher 2019 averaged Y t/ha across FEWS FNIDs (fct_yield / yield_raw_data). Not comparable to FAOSTAT national maize production (fct_production , production_grain=physical)."

PROD (national): "Nigeria maize production in 2022 was Q tonnes (agg_production_country_year , physical grain, FAOSTAT source_key ...)."

FS: "In [fnid], [month], X% of the population was in IPC Phase 3+ (FEWS population branch). Classification-only rows excluded."

EL: "Agricultural employment share for women in [country] [year] was Z% (fct_employment , unit=%, sex=female). Do not add to male share if both are percentages of the same labour force without checking the indicator definition."

GYI: "Female control of farm income in the ILRI [survey] sample was S (fct_household). This is not SDG 5.a.1 from fct_gender_inclusion."

PRC: "Retail maize price in [market] [year-month] was P local/unit (WFP). No FEWS common_unit_price on this row."

FVC / trade: "Cross-border rice flow at [border_point] in [month] used fct_trade with trade_grain=fews_border_month — not FAOSTAT country-year trade."

8. Sparse unions — how to read empty cells

Completeness is **by grain × source**, not table-wide.

Table	Expected null
fct_food_security	pct_phase* on classification rows
fct_prices	common_unit_price on WFP/FAOSTAT
fct_investment	institution/sex on credit-FDI; item on ASTI
fct_climate	unused metric columns on the other climate_grain
fct_gender_inclusion	industry/factor on land-rights items
fct_production	area/qty/yield convenience cols when element does not match

Citation keys (source_key , as_of_date , fact PK) must still be complete.

9. Out of scope

- Child anthropometry (stunting, wasting, MDD-C)
 - Labelled regenerative / CA adoption
 - Dedicated `fct_market_access` if not built
 - Catalog-only empty dims: `policy_area`, `technology`, `full_pest`, `farm_practice`
 - ILRI dairy cow-day series inside production/yield facts
-

10. Publish checklist

- ☐ Grain stated (and correct fact: production vs yield; `trade_grain`; `climate_grain`; ...)
- ☐ `source_key` resolves in `dim_source`
- ☐ `as_of_date` is observation (or basis flagged)
- ☐ Units compatible
- ☐ Discriminator applied (`production_grain`, `measure_type`, `input_grain`, ...)
- ☐ Citations = rows for evidence scoring / ACF